

# Stagehand Phase Evidence

Current checkpoint: public Site v73, verified 2026-09-03.

## Project

Stagehand is a WebMCP-native 2D animation studio for editable paper-cutout scenes. The current cut supports rigged characters, imported image props, per-asset visual direction, camera work, captions, non-voice audio cues, storyboard beats, multi-scene Preview, and deterministic WebM export.

Private source: [github.com/swaymun/stagehand-animation-studio](https://github.com/swaymun/stagehand-animation-studio)

Commit: 3f27cdbc0c7336557e9af81f6bab4a7de6bfd214

Public Site: [stagehand-animation-studio.saimun-h-shahee.chatgpt.site](https://stagehand-animation-studio.saimun-h-shahee.chatgpt.site)

Sites version: 73

## Acceptance loop

Implement - local tests - Sites deployment - hosted test - UI roast - fixes - second review.

## Implemented at this checkpoint

- Revision-safe WebMCP mutations with optional expectedRevision and structured REVISION\_CONFLICT results.
- Idempotent WebMCP mutations with optional idempotencyKey replay protection.
- Structured per-asset role, treatment, silhouette, palette, and direction notes with human and agent parity.
- Motion actions show affected character and bounded duration; pose presets remain a separate group.
- Agent frame-rate-only render updates preserve the existing resolution until a new preset is explicitly chosen.
- Storyboard mode selects the Board rail; Preview hides editing mutation surfaces and selection chrome.

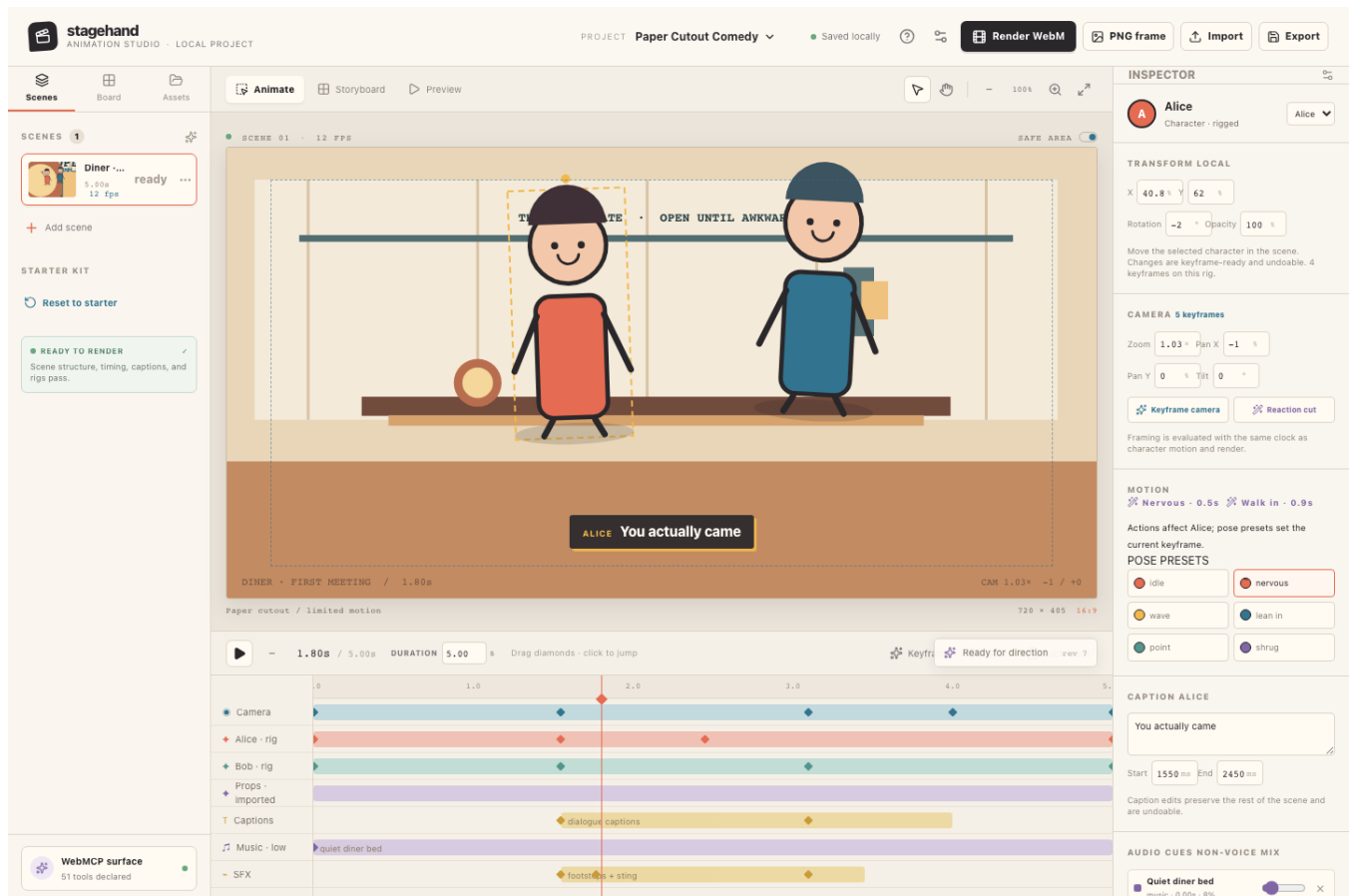
## Verification

Local gate:

```
npm run format
npm run lint
npm run build
git diff --check
npm run smoke
```

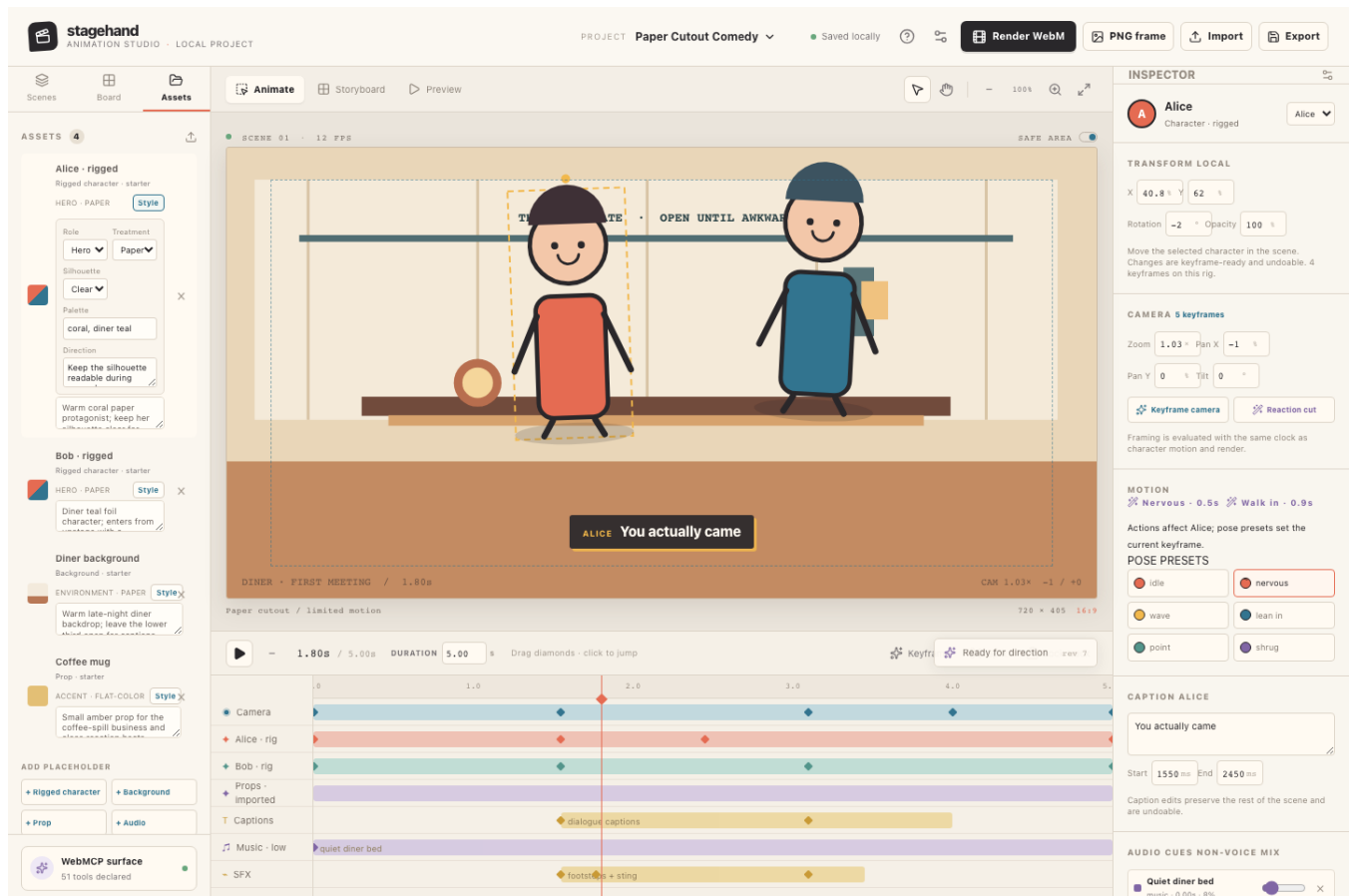
Local and hosted smoke both passed with 51 registered tools, 38 guarded mutations, stale-write conflict, idempotent replay, frame inspection, PNG frame download, asset-style parity, timeline retiming, imported-prop animation, undo/redo, a two-scene WebM download, synchronized Storyboard/Preview state, and zero page errors.

# Animate editing workspace



Hosted Playwright capture at 1440 x 960. Motion actions identify their target character and duration.

# Assets style editor



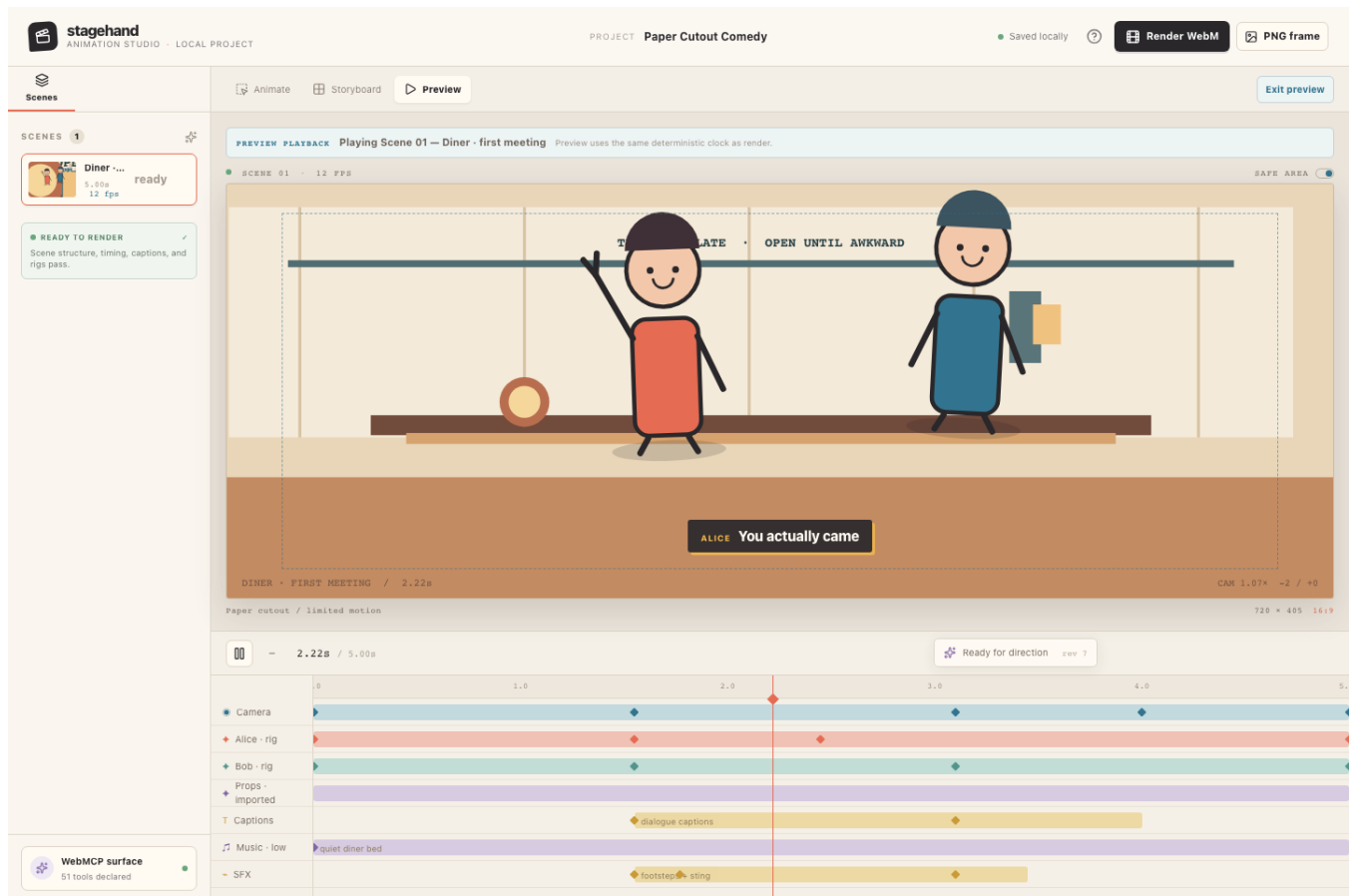
Hosted Playwright capture at 1440 x 960. The expandable Style control exposes structured visual direction without taking over the asset rail.

# Storyboard mode with Board rail

The screenshot displays the Stagehand Animation Studio interface in storyboard mode. The top bar shows the project name "Paper Cutout Comedy" and various utility buttons like "Render WebM", "PNG frame", "Import", and "Export". The left sidebar contains a "BEATS" panel with three beats: "The wait", "The entrance", and "The pause". The main workspace is titled "Block the awkward moment" and features a "BOARD RAIL" with three storyboard panels. Each panel shows a scene with two characters, Alice and Bob, and a caption. The first panel is "The wait" (0.90s - 1.55s), the second is "The entrance" (1.55s - 3.10s), and the third is "The pause" (3.10s - 5.00s). The second panel is highlighted with a red border. Below the storyboard panels is a timeline with a playhead at 1.80s. The timeline includes tracks for Camera, Alice - rig, Bob - rig, Props - imported, Captions, Music - low, and SFX. The right sidebar contains an "INSPECTOR" panel for the selected character "Alice", showing transform and camera properties. The bottom of the interface shows a "WebMCP surface" with 51 tools declared.

Hosted Playwright capture at 1440 x 960. The selected top-level mode and project rail stay synchronized.

# Review-first Preview player



Hosted Playwright capture at 1440 x 960. Scene-only context, transport, and Exit preview remain; global editing controls are hidden.

## Roast findings and fixes

- Imported props lacked direct human transform editing. Added X/Y/Scale/Rot controls sharing the keyframe command path.
- Four prop transform fields initially collided with the asset-list flex rule and truncated. Replaced it with an explicit two-column grid; computed field boxes are 57 px wide in two rows.
- Preview carried too much editing chrome. Converted it into a review-first player with scene-only context and Exit preview.
- Timeline diamonds had 7 x 7 px hit targets. Enlarged the buttons to 22 x 22 pixels while preserving the visual diamond.
- Agent retries could duplicate successful writes. Added optional idempotency-key replay protection and a hosted smoke assertion.
- Switching top-level Storyboard mode could leave Assets selected. Synced mode selection to its companion project rail.
- Character transform inputs lacked explicit accessible names. Added dynamic X/Y/rotation/opacity labels and a smoke check.
- Asset briefs carried style intent only as prose. Added structured role, treatment, silhouette, palette, and direction fields.
- Motion action buttons were visually mixed with pose presets. Added explicit durations, affected-character copy, and a Pose presets label.
- Agent frame-rate-only render updates reset an existing 1080p choice. Preserved the current resolution when the preset is omitted and added smoke coverage.
- Preview still exposed global project-editing controls. Hid Settings, Import, Export JSON, and rename while retaining WebM and PNG review actions.

## Limits

Direct CUA inspection was unavailable because the Mac was locked; Playwright was used for hosted interaction and screenshots. The smoke harness proves injected modelContext registration and execution, not live production ChatGPT WebMCP discovery. TTS, voice cloning, dialogue recording, phoneme extraction, and lip-sync remain out of scope.